



Group 2 – CECS1010

WATER QUALITY LIVE MAP

OUTLINE

Overview

Software
Description

Hardware
Description

Conclusions

CURRENT PROBLEMS

Public awareness



Water-side areas



Overview

- Purpose:
 - Create a real-time mapping of water quality
 - Measure water quality in distant area where people can't normally reach
 - Store and display data in a server/ website
 - Raise people awareness about local water quality
 - Early warning a bout possible water pollution

Flow graph



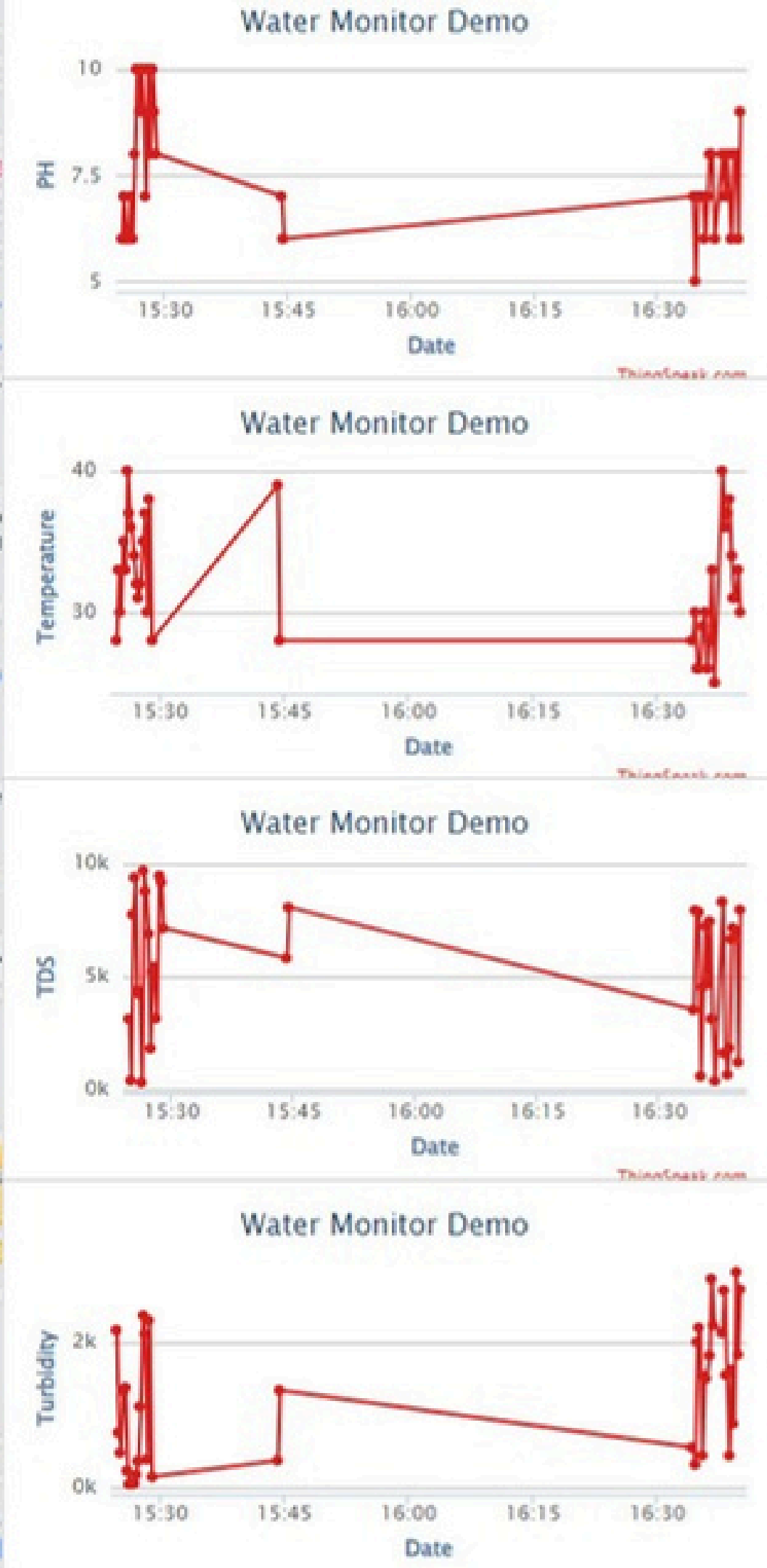
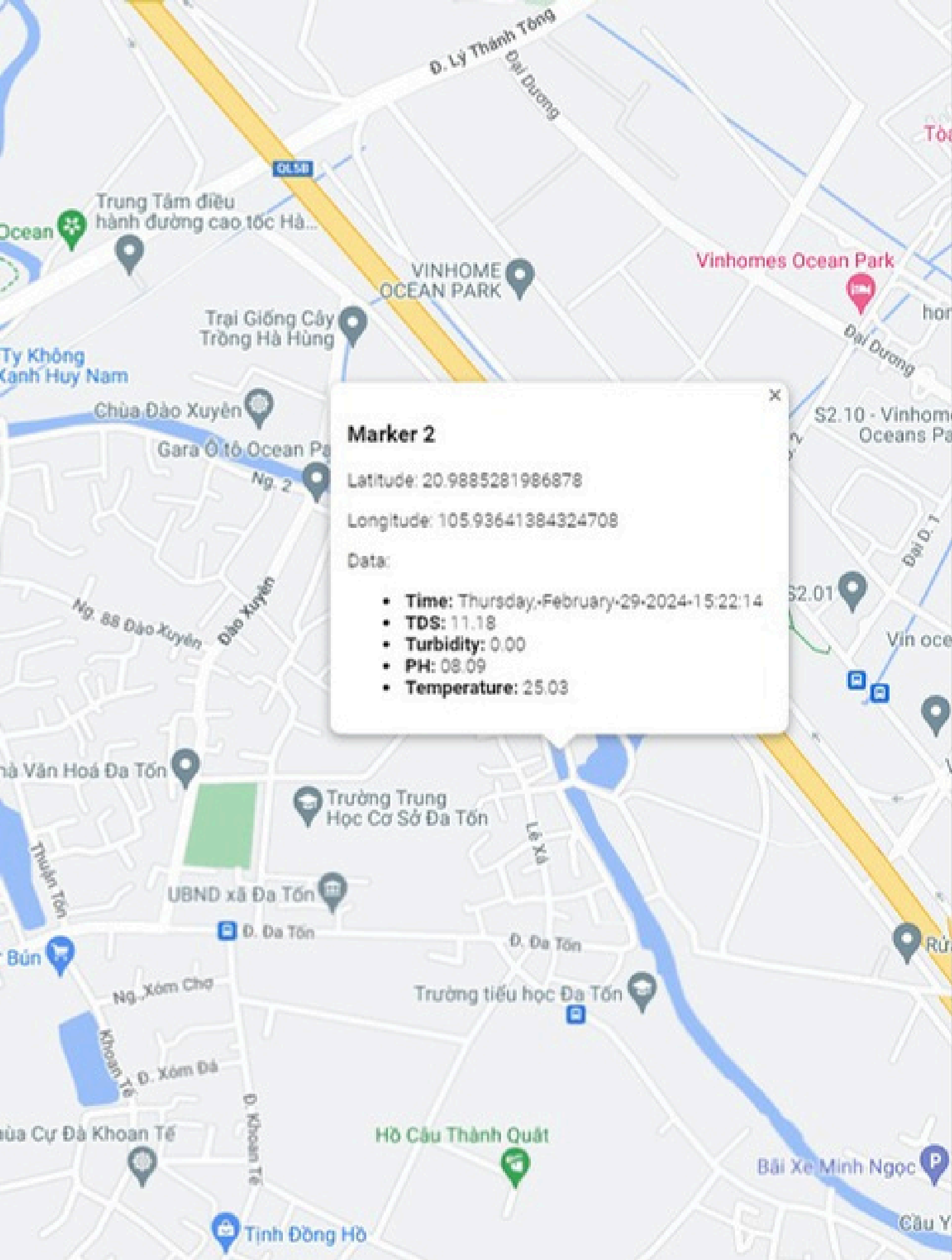
**Sensor
block**



Database

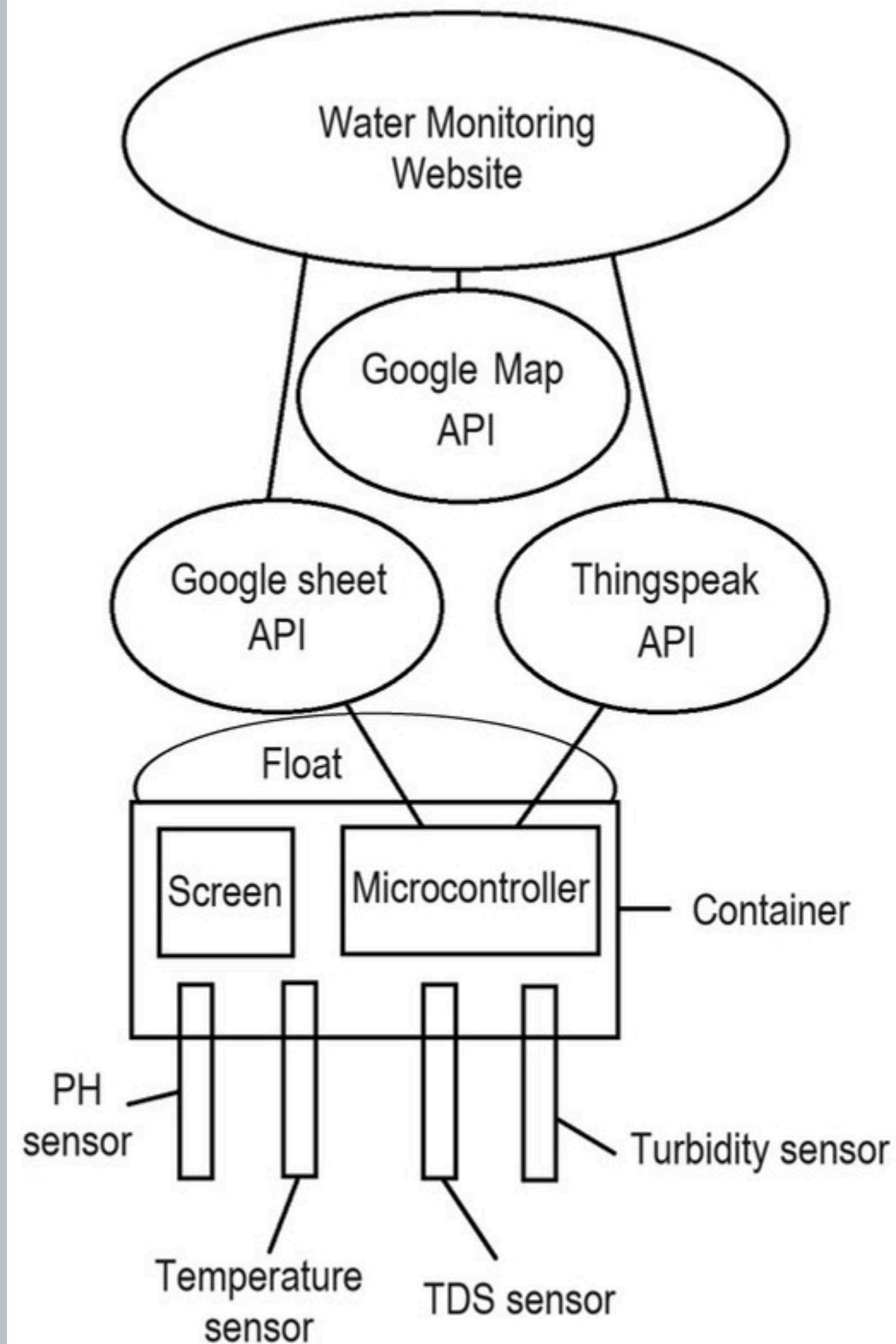


**Graph
plot**



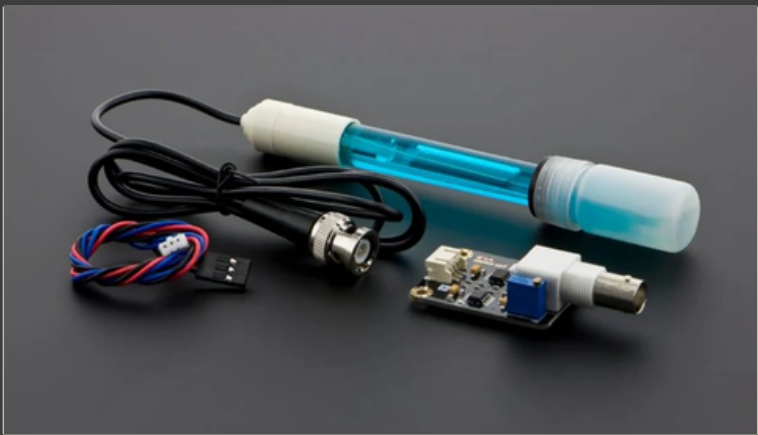
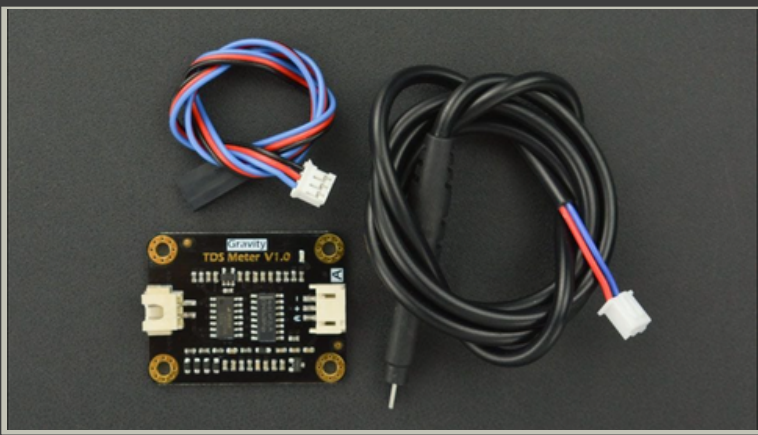
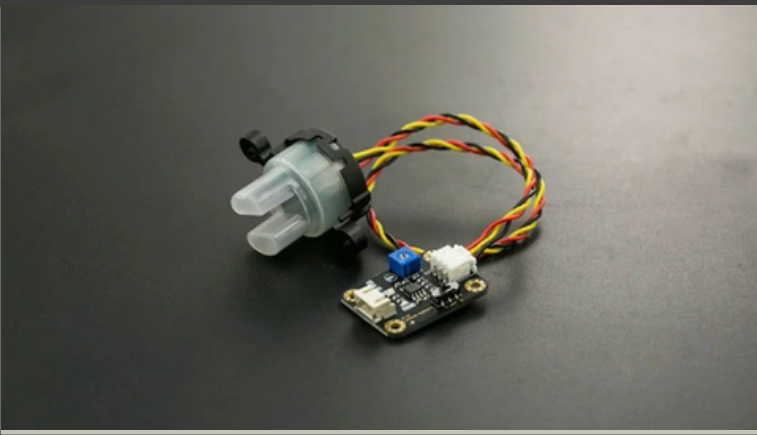
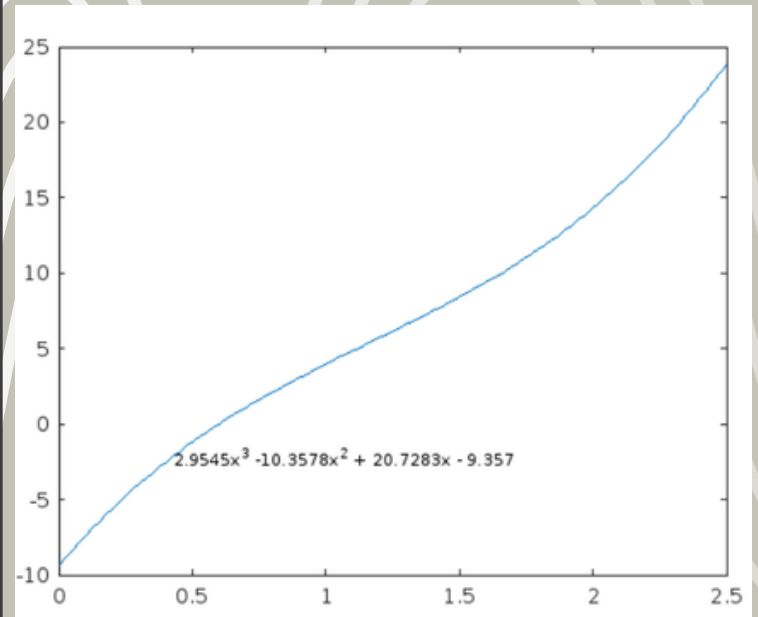
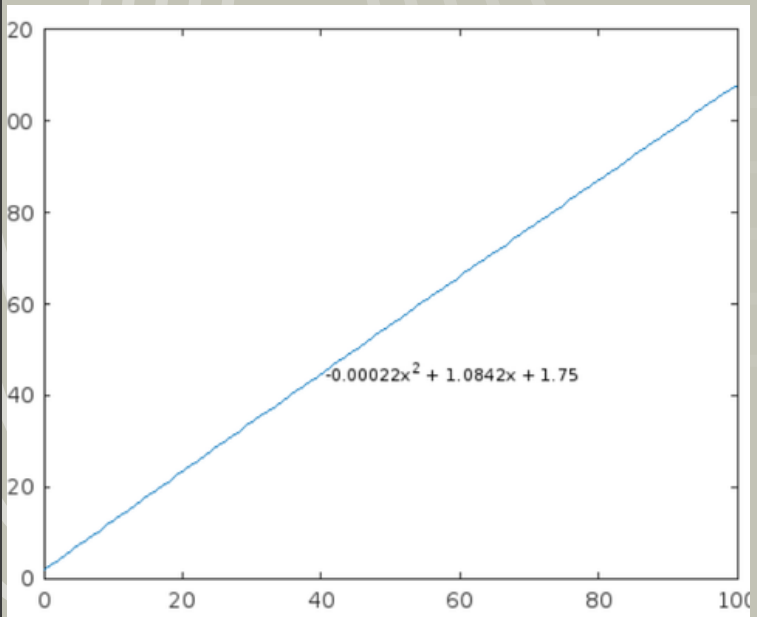
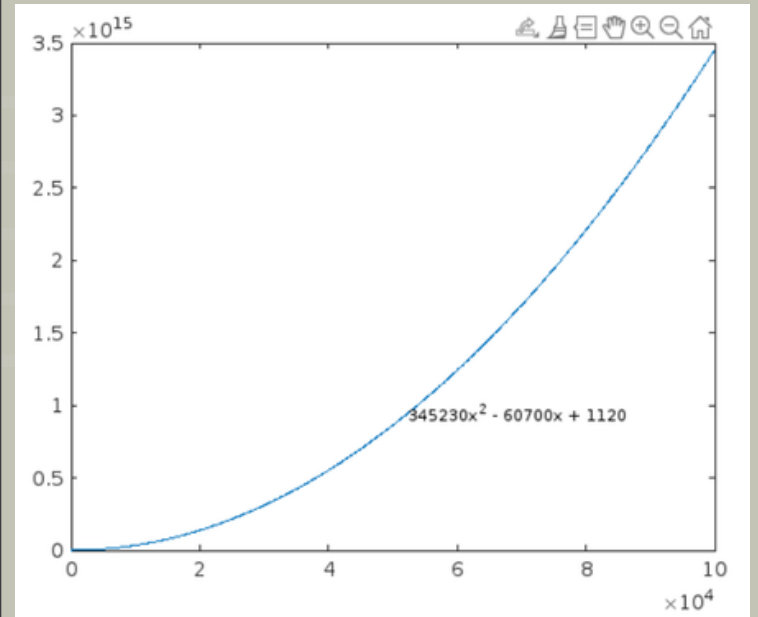
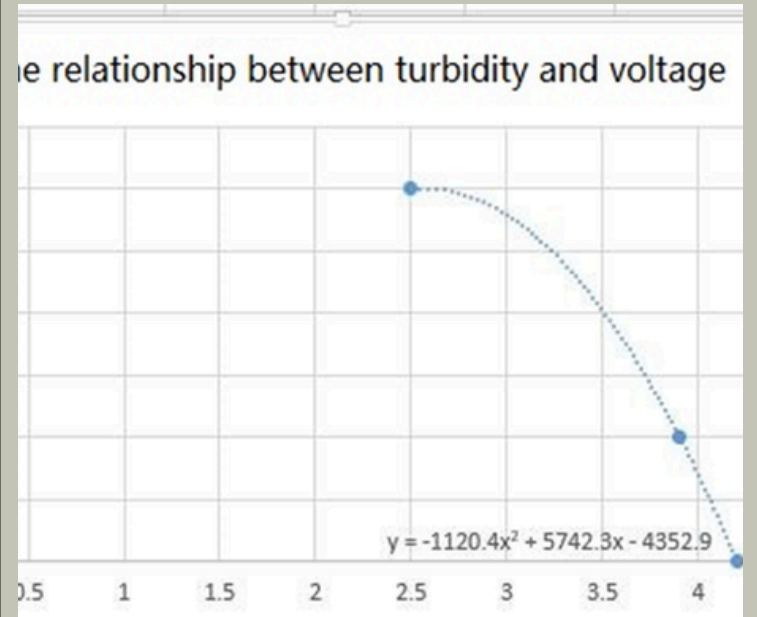
Web
Display

HIGH LEVEL DIAGRAM

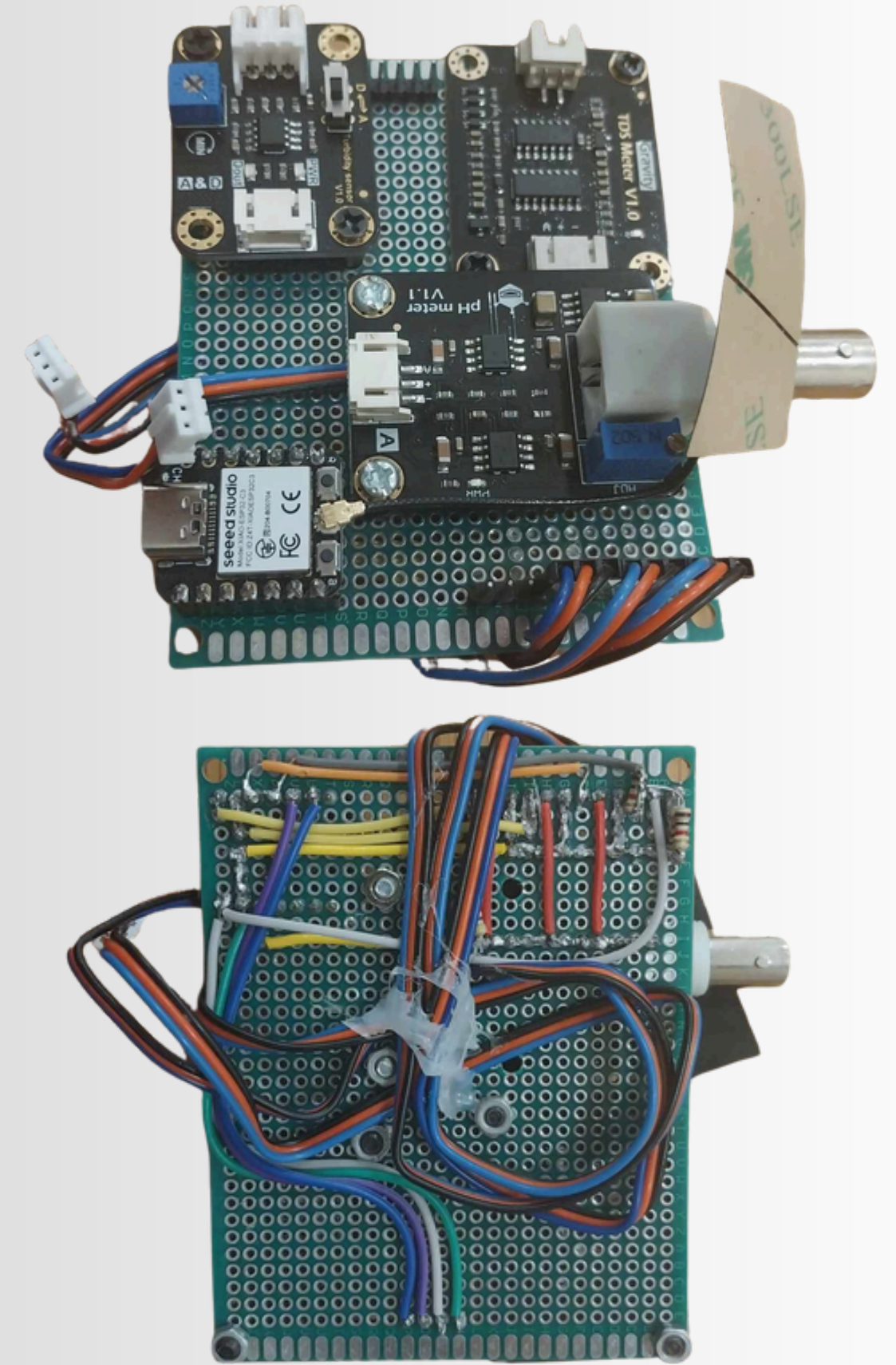
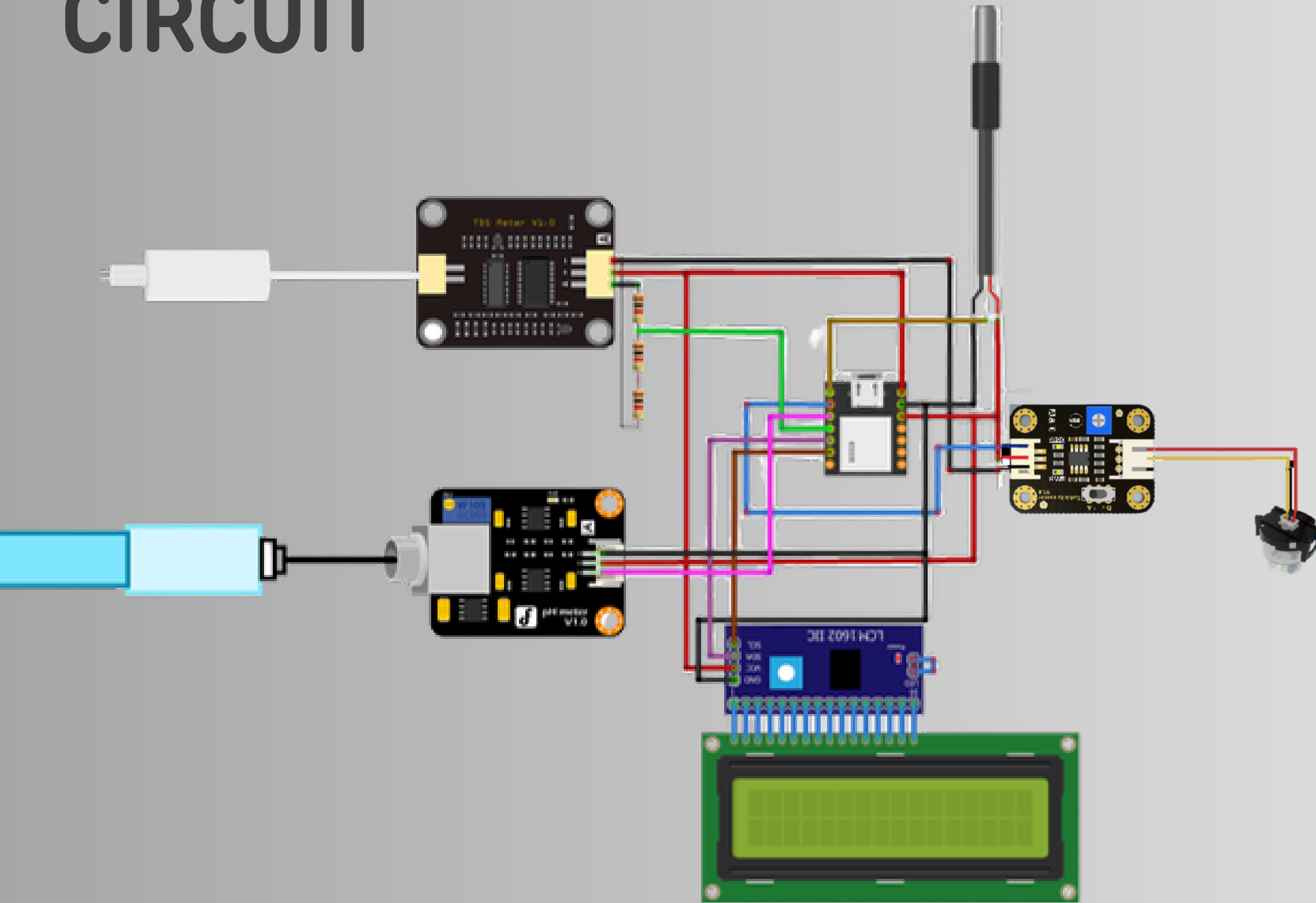


Hardware description

SENSORS

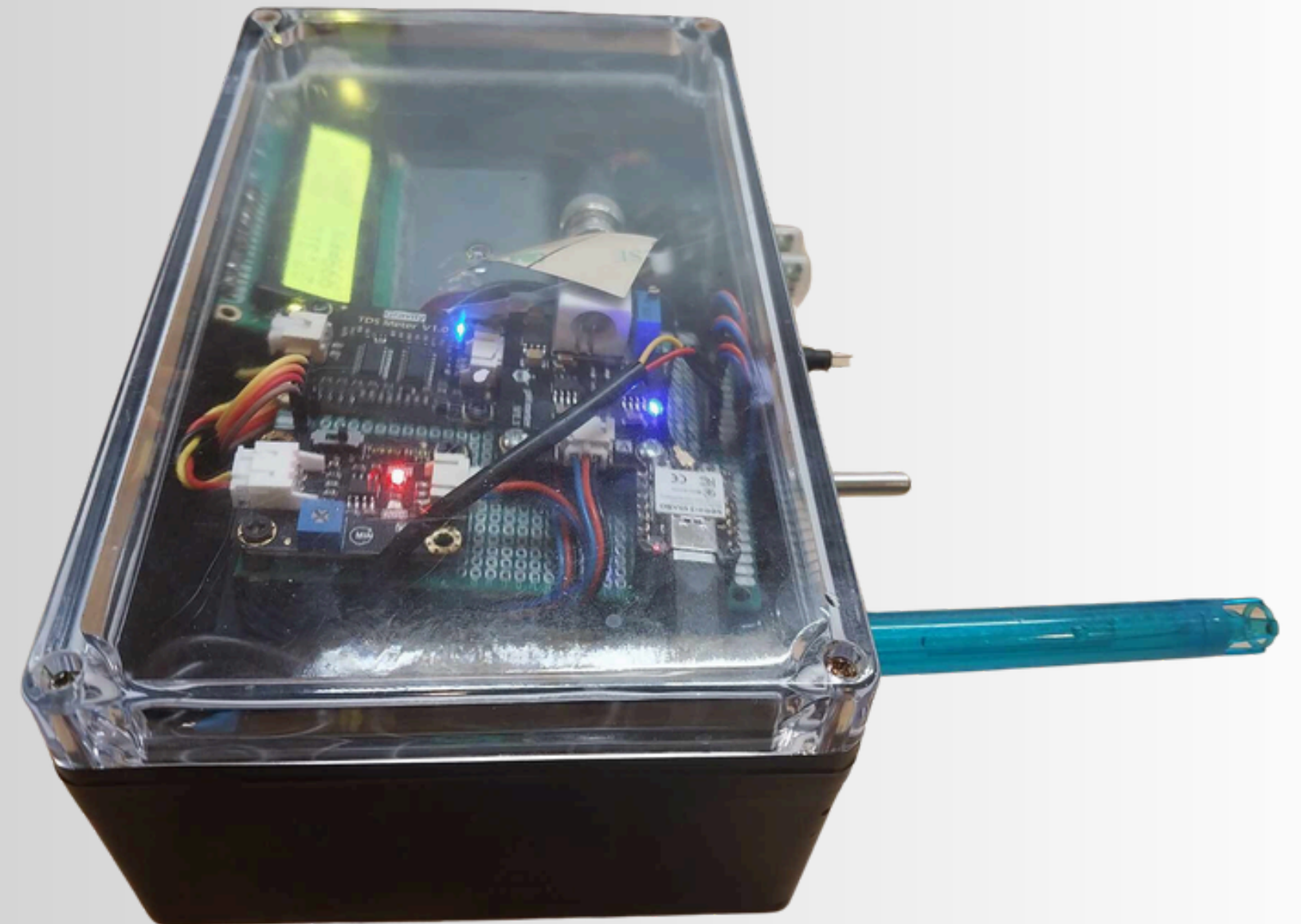
				
Uncertainty	+ 0.6 / -0.6	+0.36 / -0.36	+91 / -91	+3 / -3
Response Time	2 minutes	2 minutes	20 seconds	20 seconds
Calibrated graph				

CIRCUIT



The whole circuit is soldered onto a green strip board and each module is detachable from the board

CONTAINER



The waterproof technical box is machined with gaps for the sensor, sealed with APOLLO silicone glue and waterproof tape, transparent lid for the display

Software description

Extracting the data
from sensors

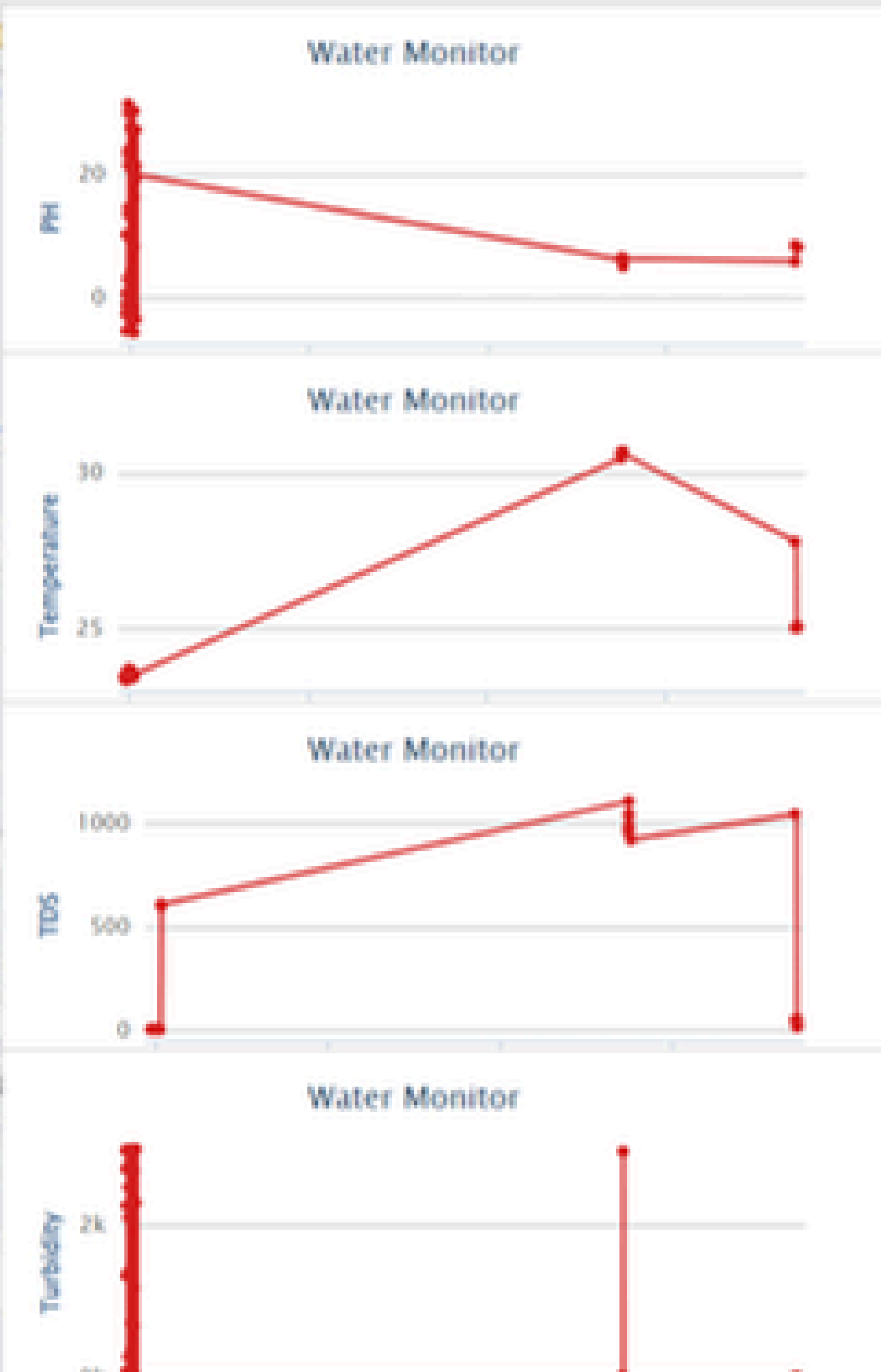
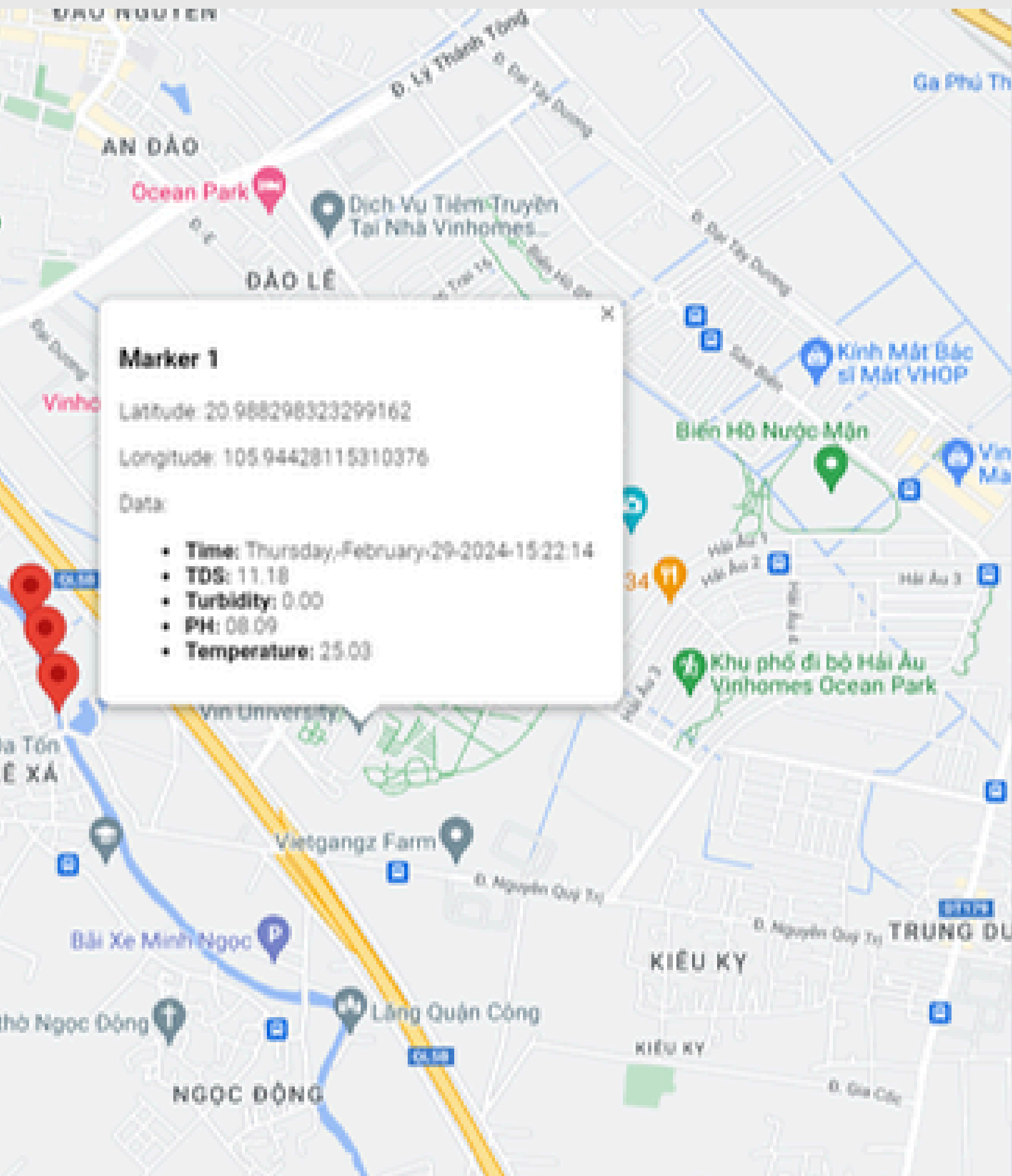
```
float PH = GetPHCalibrated(getVoltageFilterAvg(20, 4));
float TDS = GetTDSCalibrated(getVoltageFilterAvg(20, 5));
float Turb = GetTurbCalibrated(getVoltageFilterAvg(20, 3));
float Temp = GetTempCCalibrated(ds.getTempC());
```

	A	B	C	D	E
1	Time	TDS	Turbidity	PH	Temperature
2	Monday,-January-22-2024-12:09:56	1.93	0	1.54	25.03
3	Monday,-January-22-2024-12:10:07	2.21	0	1.55	24.96
4	Monday,-January-22-2024-12:10:18	2.34	0	1.55	24.89
5	Monday,-January-22-2024-12:10:51	1.72	0	1.54	24.82
6	Monday,-January-22-2024-12:11:03	2.17	0	1.55	24.82
7	Monday,-January-22-2024-12:11:14	2.21	0	1.55	24.82
8	Monday,-January-22-2024-12:11:32	1.9	0	1.54	24.89
9	Monday,-January-22-2024-12:11:44	2.07	0	1.54	24.89
10	Monday,-January-22-2024-12:11:56	2.24	0	1.54	24.89
11	Monday,-January-22-2024-06:42:06	1.93	0	1.55	24.96
12	Monday,-January-22-2024-06:42:17	2.03	0	1.55	24.96
13	Monday,-January-22-2024-06:42:29	2.31	0	1.55	25.03
14	Monday,-January-22-2024-06:42:45	2.1	0	1.54	24.96
15	Monday,-January-22-2024-06:42:57	2.24	0	1.55	24.96
16	Monday,-January-22-2024-06:43:08	2.41	0	1.55	24.96
17	Monday,-January-22-2024-06:43:19	2.17	0	1.55	24.96
18	Monday,-January-22-2024-06:43:30	2.45	0	1.55	25.03
19	Monday,-January-22-2024-06:43:42	2.28	0	1.55	25.03
20	Monday,-January-22-2024-06:43:53	2.31	0	1.55	24.96
21	Monday,-January-22-2024-06:44:04	2.21	0	1.55	25.03
22	Monday,-January-22-2024-06:44:16	2.38	0	1.55	24.96
23	Monday,-January-22-2024-06:44:27	2.17	0	1.55	25.03
24	Monday,-January-22-2024-06:44:39	2.55	0	1.55	27.58
25	Monday,-January-22-2024-06:44:50	2.21	0	1.55	32.2

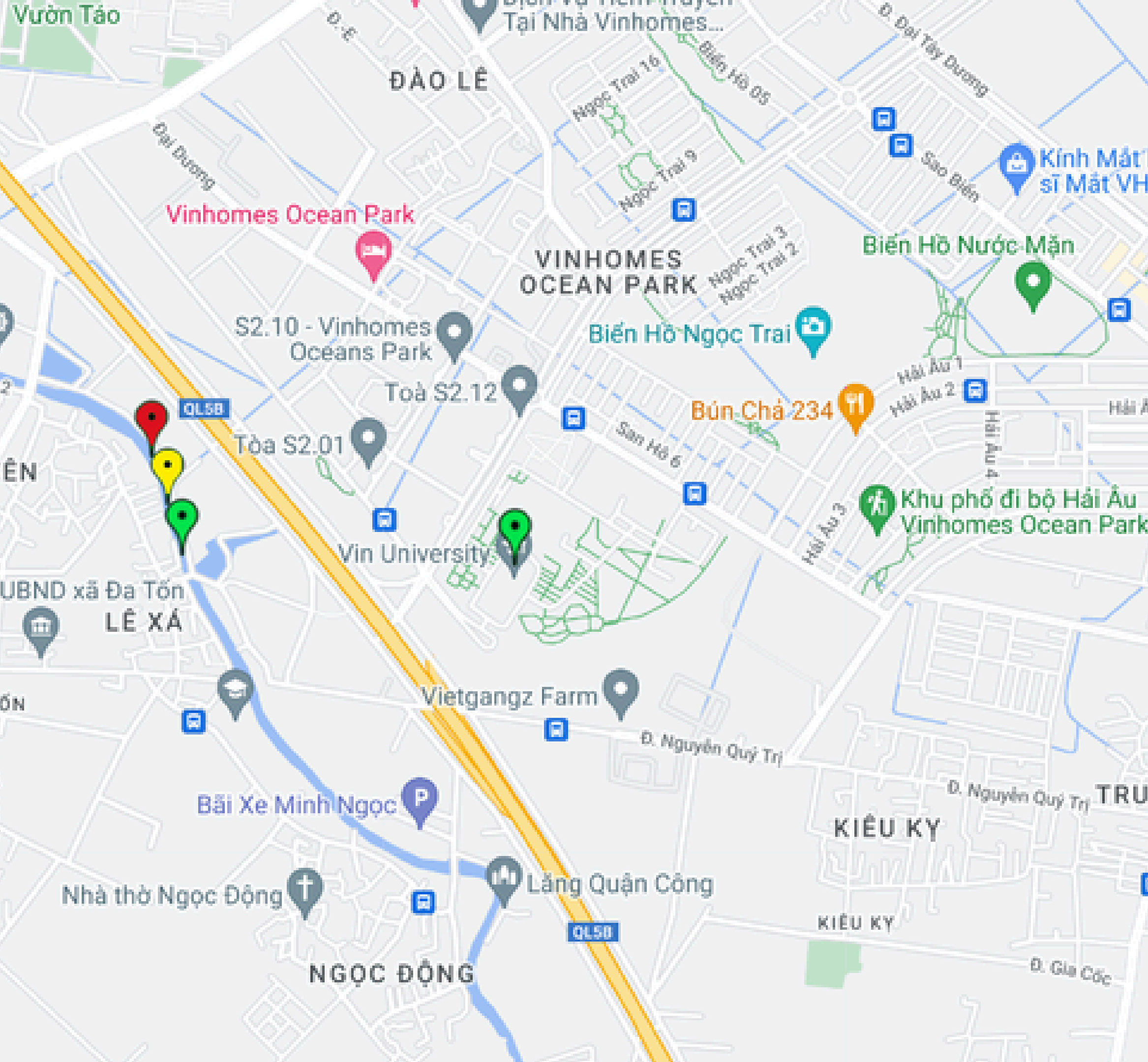
```
valueRange.add("majorDimension", "COLUMNS");
valueRange.set("values/[0]/[0]", asString);
valueRange.set("values/[1]/[0]", String(TDS));
valueRange.set("values/[2]/[0]", String(Turb));
valueRange.set("values/[3]/[0]", String(PH));
valueRange.set("values/[4]/[0]", String(Temp));

success = GSheet.values.append(&response, spsID, "Sheet1!A1", &valueRange);
```

Sending the Data to the Database



Real
time
graph
+
Detailed
data



Color marking

Green: Safe

Yellow: Harmful

Red: Dangerous



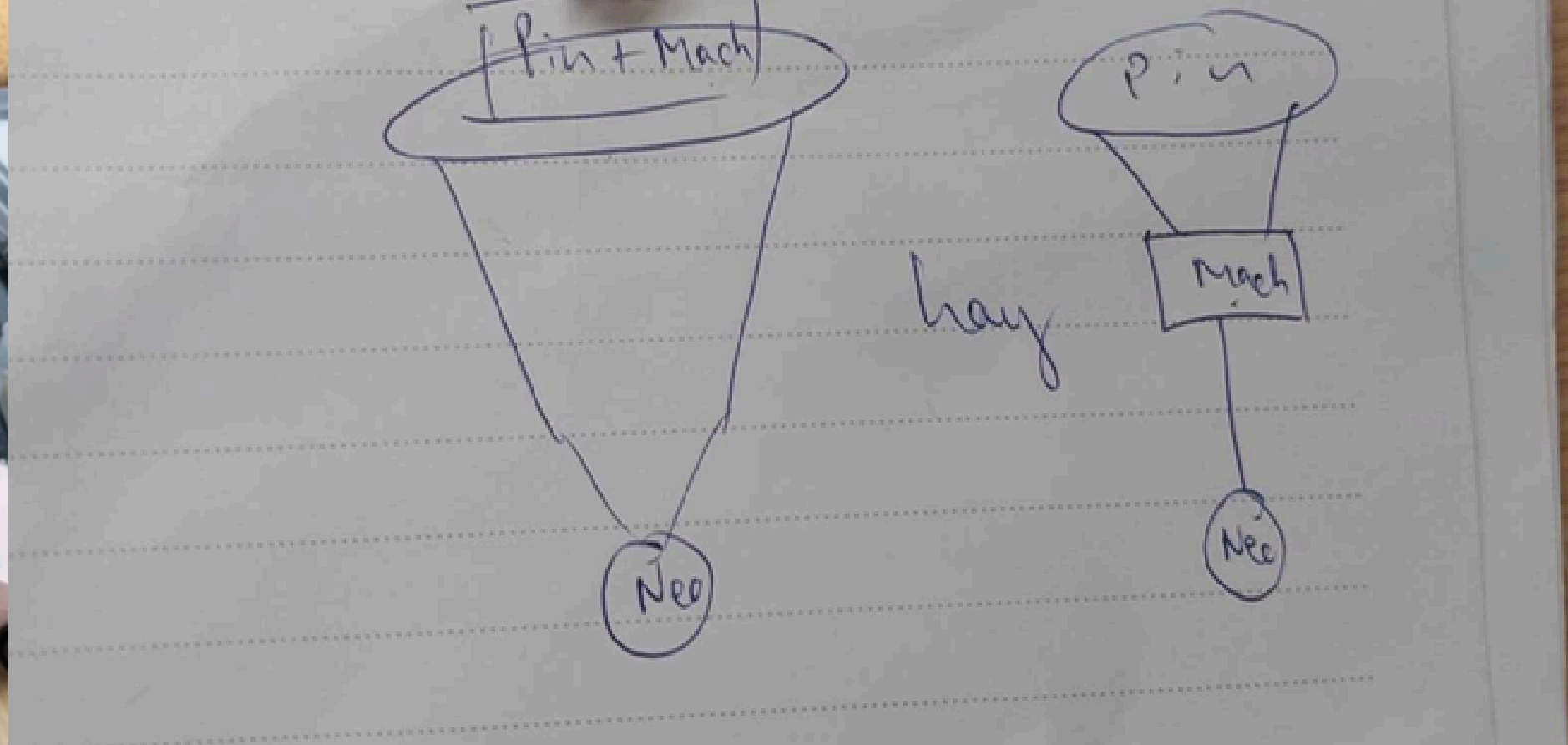
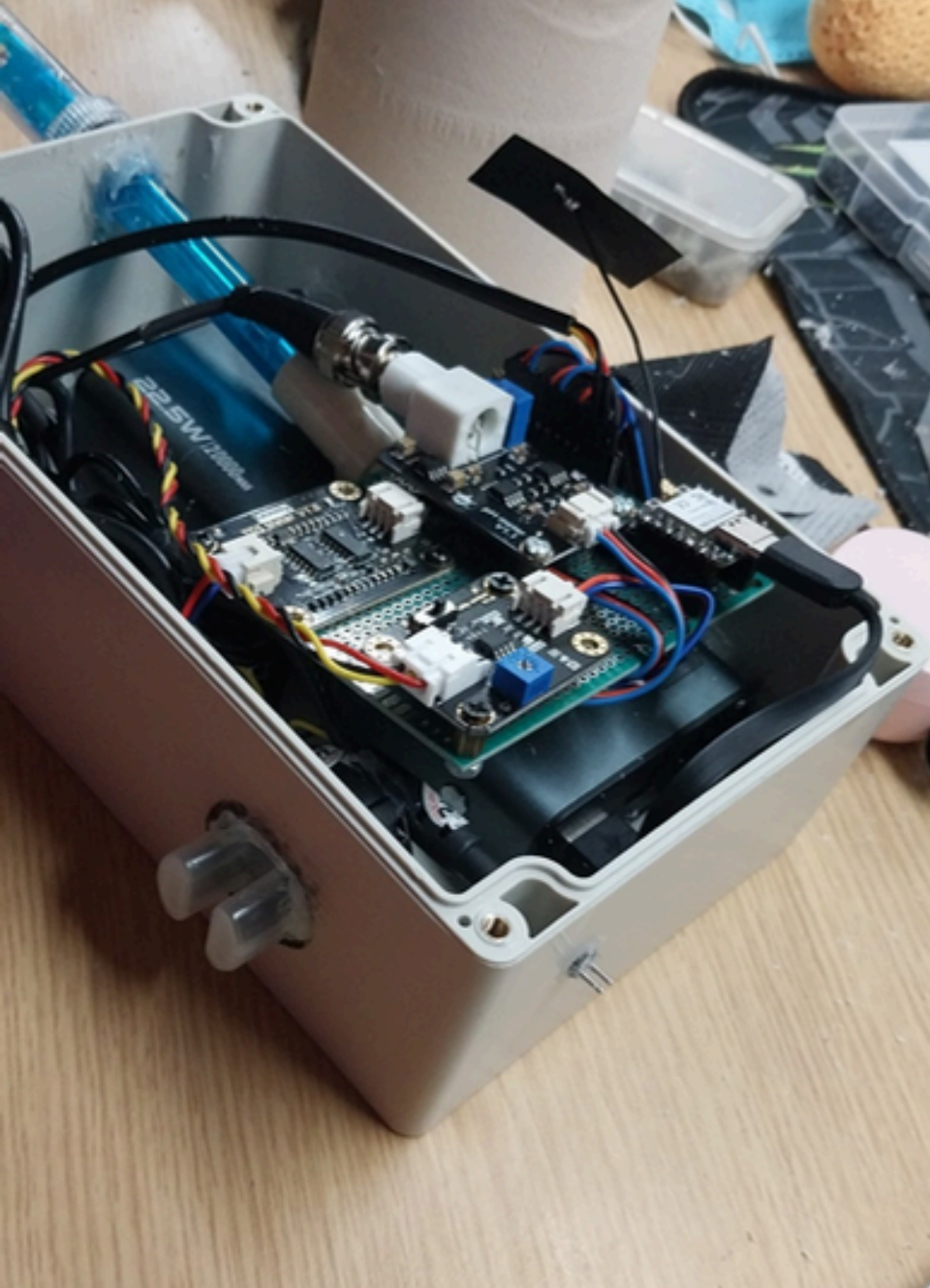
CONCLUSIONS

Pros

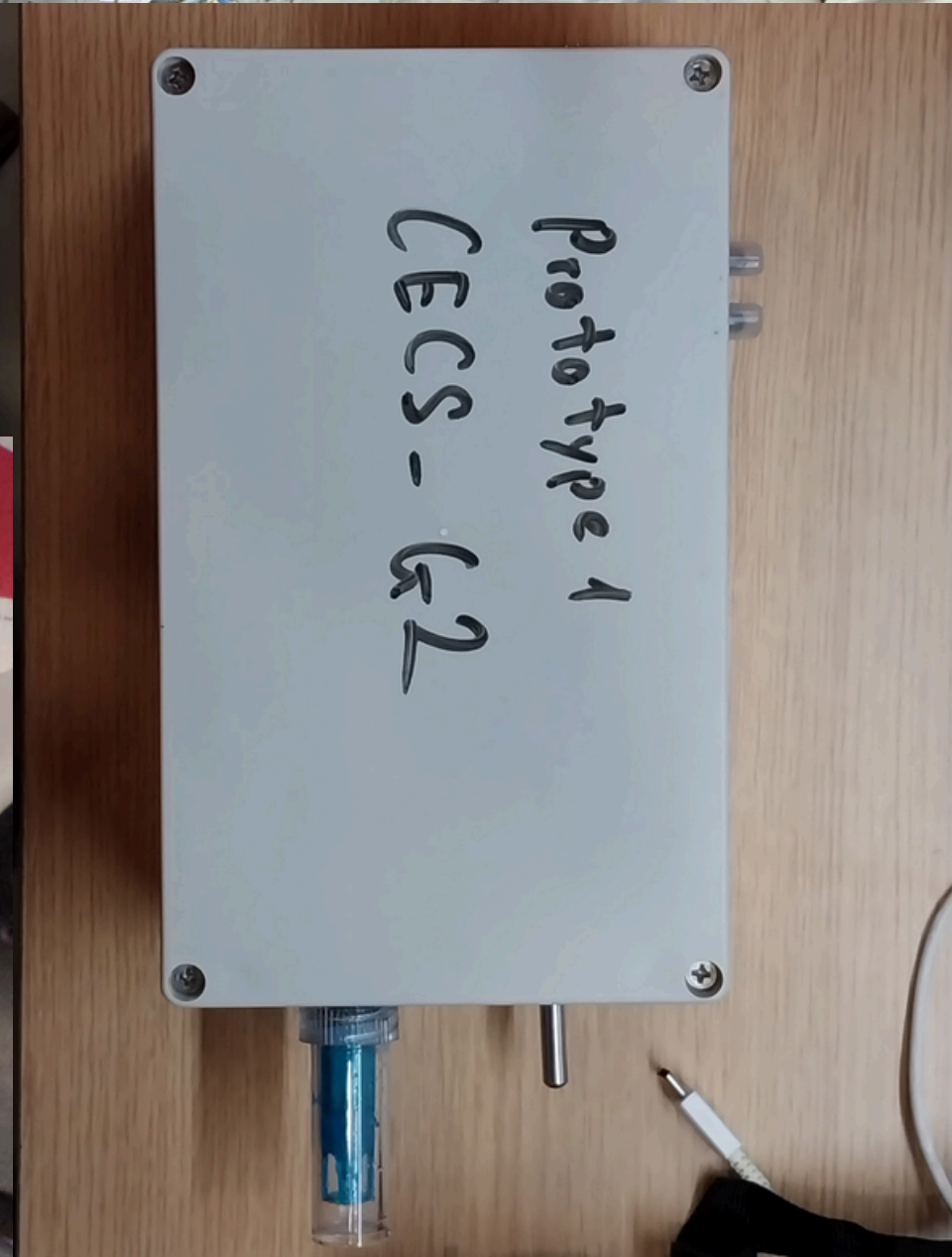
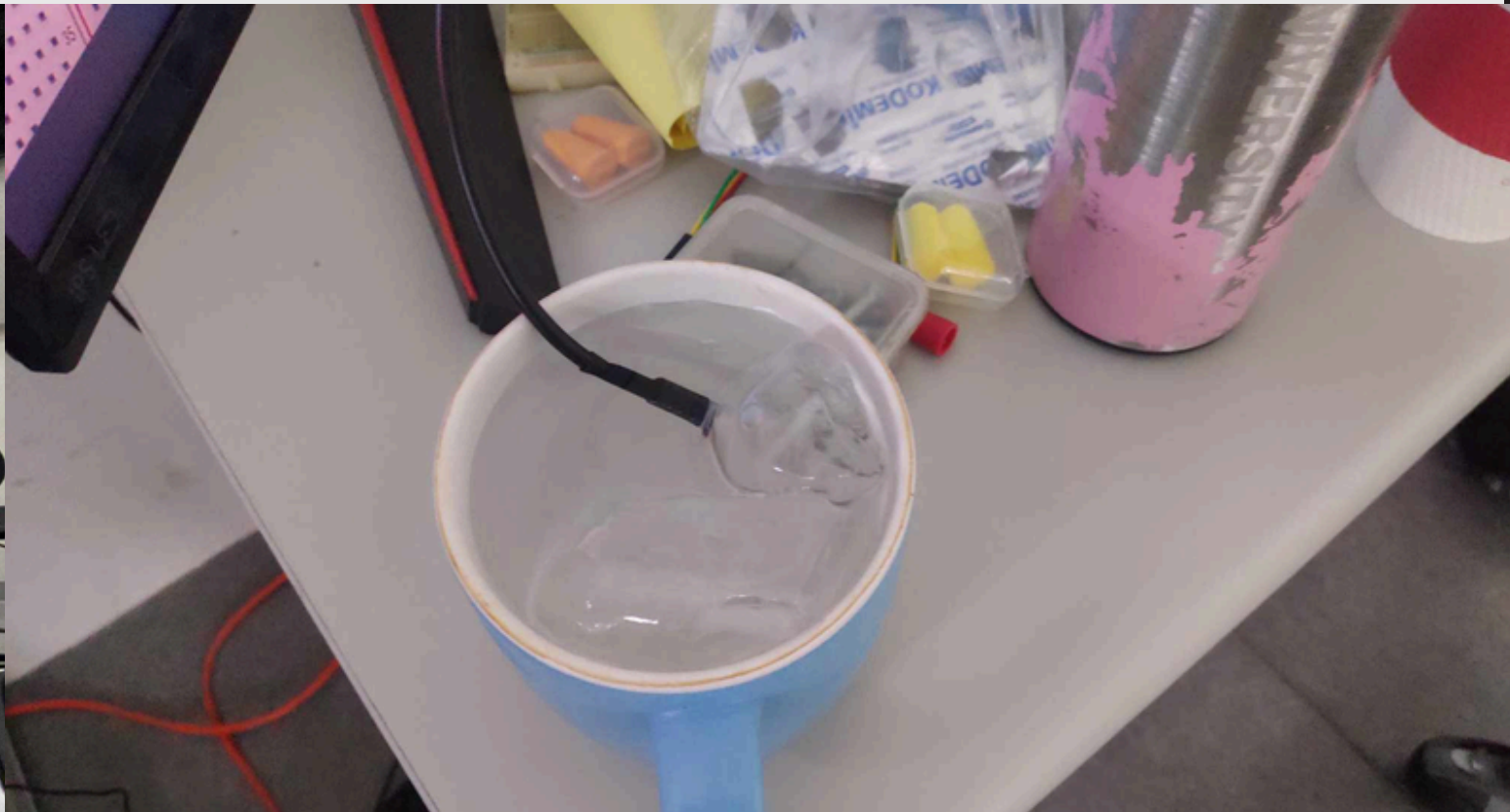
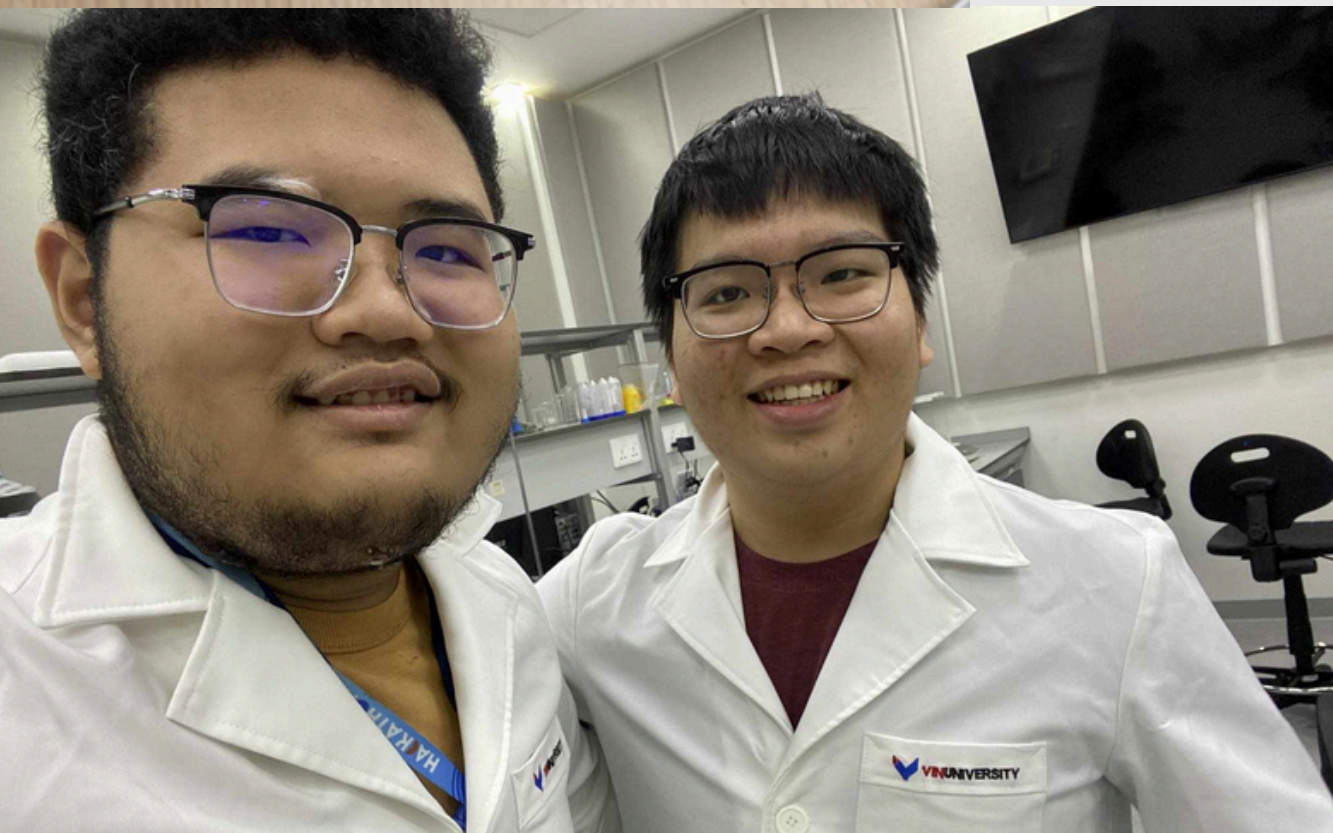
- Provide a qualified database for multi-purpose usage.
- Equipped with advanced sensors and data analysis tools to provide real-time updates
- Integration web applications for remote data update
- Utilize IoT for real-time data collection even from remote locations.
- Develop a user-friendly interface
- Collaborate with local authorities and communities to increase the covered regions.

Cons

- Low usage time based on battery
- Uncertainty still high on some sensors
- Hard to calibrate, maintain
- There will be a report feature



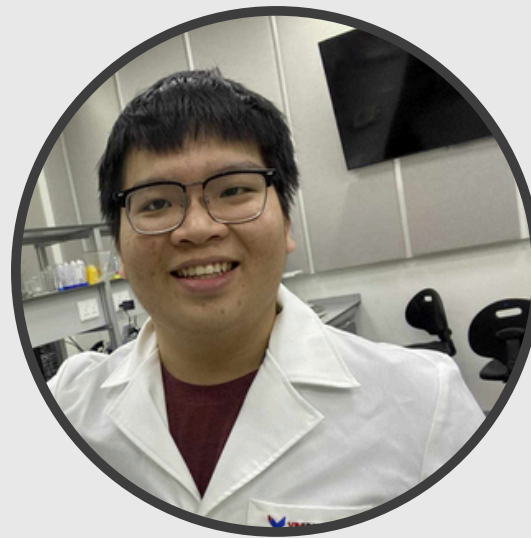
THE PROCESS



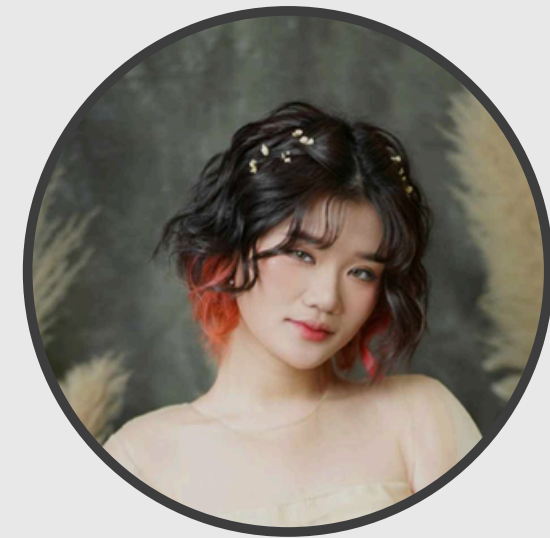
THANK YOU



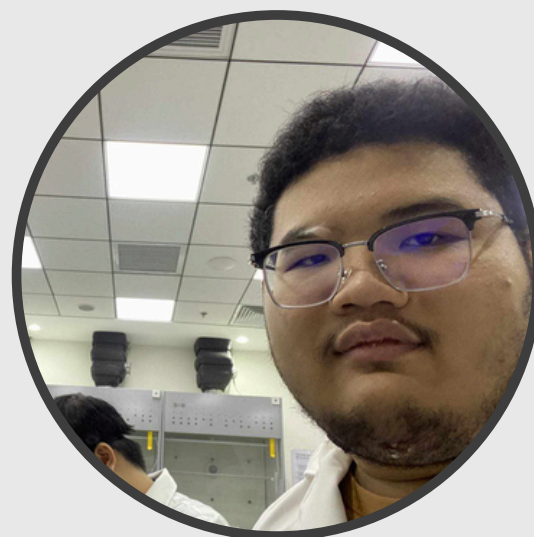
Time Keeper-Hai Trung



Facilitator-Nghiep Quy



Note Taker-Anh Thu



Reporter-Minh Khang



Member-Quoc Cuong